

Asset Guidelines

This document provides specifications for creating or sourcing assets for the Space Shooter game.

Sprite Specifications

General Guidelines

- **Format:** PNG with transparency
 - **Color Mode:** RGBA
 - **Pixels Per Unit:** 100 (default Unity setting)
 - **Filter Mode:** Point (for pixel art) or Bilinear (for smooth graphics)
 - **Compression:** None or Low Quality for best results
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Player Assets

Player Ship (`player_ship.png`)

- **Size:** 64x64 pixels or 128x128 pixels
- **Design:** Triangle or classic spaceship shape pointing upward
- **Colors:** Blue/cyan primary with white/silver accents
- **Style:** Your choice - pixel art or smooth vector style

Simple Placeholder:

Create a 64x64 image with:

- Isosceles triangle pointing up
- Fill: #00AAFF (cyan blue)
- Outline: #FFFFFF (white), 2px

Player Bullet (`bullet_player.png`)

- **Size:** 8x16 pixels or 16x32 pixels
 - **Design:** Elongated oval or energy bolt
 - **Colors:** Bright cyan (#00FFFF) with white center
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Enemy Assets

Basic Enemy (`enemy_basic.png`)

- **Size:** 48x48 pixels or 64x64 pixels
- **Design:** Simple angular shape (square with cut corners, hexagon)
- **Colors:** Red (#FF4444) with darker red outline

Zigzag Enemy (`enemy_zigzag.png`)

- **Size:** 48x48 pixels
- **Design:** Diamond or arrow shape

- **Colors:** Orange (#FF8800)

Circular Enemy (enemy_circular.png)

- **Size:** 48x48 pixels
- **Design:** Circular or ring shape
- **Colors:** Purple (#AA44FF)

Charger Enemy (enemy_charger.png)

- **Size:** 48x48 pixels
- **Design:** Pointed/aggressive shape
- **Colors:** Yellow (#FFFF44)

Boss Enemy (enemy_boss.png)

- **Size:** 128x128 pixels or 192x192 pixels
- **Design:** Large imposing ship, more detailed
- **Colors:** Dark red (#880000) with glowing accents

Enemy Bullet (bullet_enemy.png)

- **Size:** 12x12 pixels
 - **Design:** Small circle or diamond
 - **Colors:** Red (#FF3333) with orange glow
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Power-Up Assets

Generic Power-Up (powerup.png)

- **Size:** 32x32 pixels
- **Design:** Diamond, star, or capsule shape
- **Note:** Color is set via script, so use white base

Shield Power-Up Icon

- **Color:** Cyan (#00DDFF)
- **Symbol:** Shield outline

Rapid Fire Power-Up Icon

- **Color:** Yellow (#FFDD00)
- **Symbol:** Lightning bolt or triple arrows

Health Power-Up Icon

- **Color:** Green (#44FF44)
- **Symbol:** Plus sign or heart

Extra Life Power-Up Icon

- **Color:** Magenta (#FF44FF)
 - **Symbol:** Small ship or 1UP text
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Background Assets

Space Background (`background_space.png`)

- **Size:** 1920x1080 pixels or tiling texture (512x512)
- **Design:** Dark blue/black gradient with distant stars
- **Colors:**
 - Top: Dark blue (#0a0a2a)
 - Bottom: Near black (#050510)
 - Stars: White dots of varying sizes

Nebula Overlay (`background_nebula.png`)

- **Size:** 512x512 pixels (tiling)
 - **Design:** Semi-transparent colorful clouds
 - **Colors:** Purple, blue, pink hues at low opacity (20-40%)
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UI Assets

Button Backgrounds

- **Size:** 200x50 pixels (stretchable)
- **Design:** Rounded rectangle or sci-fi panel style
- **Colors:** Dark blue (#1a1a3a) with bright border (#00AAFF)

Health Bar

- **Background:** Dark gray (#333333)
- **Fill:** Gradient from green to red based on health
- **Border:** White or cyan

Panel Backgrounds

- **Style:** Semi-transparent dark panels
 - **Colors:** #000000 at 80% opacity with colored border
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Creating Placeholder Sprites

If you don't have art assets, you can create simple placeholders:

Using Unity's Built-in Shapes:

1. In Project window: Right-click > Create > 2D > Sprites
2. Choose Circle, Square, Triangle, etc.
3. Tint via SpriteRenderer color property

Using Code-Generated Sprites:

```
// Create a simple colored square sprite
public static Sprite CreateSquareSprite(int size, Color color)
{
    Texture2D texture = new Texture2D(size, size);
    Color[] pixels = new Color[size * size];
    for (int i = 0; i < pixels.Length; i++)
        pixels[i] = color;
    texture.SetPixels(pixels);
    texture.Apply();
    return Sprite.Create(texture, new Rect(0, 0, size, size), new Vector2(0.5f, 0.5f))
;
}
```

Free Asset Sources

Here are some free asset sources you can use:

1. **OpenGameArt.org** - Free game assets (check licenses)
2. **Kenney.nl** - High-quality free game assets (CC0)
3. **itch.io** - Many free sprite packs
4. **Unity Asset Store** - Search for free 2D space assets

Recommended Free Packs:

- Kenney's "Space Shooter Redux" (free, CC0)
- Kenney's "Space Shooter Extension" (free, CC0)

Audio Specifications

Sound Effects

- **Format:** WAV or OGG
- **Sample Rate:** 44100 Hz
- **Channels:** Mono (for SFX) or Stereo
- **Duration:** 0.1s - 2s for most effects

Required Sound Effects:

Name	Description	Duration
PlayerShoot	Laser/pew sound	0.1-0.3s
PlayerHit	Impact/damage	0.2-0.5s
PlayerDeath	Explosion	0.5-1s
EnemyShoot	Enemy laser	0.1-0.3s
EnemyDeath	Small explosion	0.3-0.5s
BossSpawn	Warning/alarm	1-2s
BossDeath	Large explosion	1-2s
PowerUp	Pickup chime	0.3-0.5s
WaveStart	Alert/notification	0.5-1s
WaveComplete	Victory fanfare	1-2s

Music

- **Format:** OGG (recommended) or MP3
- **Duration:** 2-4 minutes, loopable
- **Style:** Electronic, synthwave, or orchestral space themes

Free Audio Sources:

- **Freesound.org** - Sound effects (check licenses)
- **OpenGameArt.org** - Music and SFX
- **Incompetech.com** - Royalty-free music by Kevin MacLeod

Font Specifications

Recommended Fonts:

- **Sci-Fi Style:** Orbitron, Audiowide, Exo
- **Pixel Art:** Press Start 2P, VT323, Silkscreen

Usage:

1. Download TTF/OTF font files
2. Import into Assets/Fonts/
3. Create TextMeshPro Font Asset:
 - Window > TextMeshPro > Font Asset Creator
 - Drag font, generate atlas

Layer Setup

Configure these sorting layers in order (bottom to top):

1. Background
2. Stars
3. Projectiles
4. Pickups
5. Characters
6. Effects
7. UI

Physics Layers

Configure these physics layers:

- Default
- Player
- Enemies
- PlayerBullets
- EnemyBullets
- PowerUps

Layer Collision Matrix:

	Player	Enemies	PlayerBul-lets	EnemyBul-lets	PowerUps
Player	-	✓	-	✓	✓
Enemies	✓	-	✓	-	-
PlayerBullets	-	✓	-	-	-
EnemyBullets	✓	-	-	-	-
PowerUps	✓	-	-	-	-